

5 Data Science (djw1005)

- (a) We have two groups of paired data, (a_j, b_j) for $j \in \{1, \dots, m\}$ and (a'_k, b'_k) for $k \in \{1, \dots, m'\}$. Consider the probability model

$$B_j \sim \text{Poisson}(\lambda a_j), \quad B'_k \sim \text{Poisson}(\mu a'_k)$$

where λ and μ are unknown parameters.

- (i) Find maximum likelihood estimates for λ and μ . (ii) Describe how to test the hypothesis that $\lambda = \mu$. [5 marks]
- (b) We have a dataset in which each datapoint $i \in \{1, \dots, n\}$ consists of an integer response y_i and a collection of m non-negative features $(e_{1,i}, \dots, e_{m,i})$. A *Poisson-linear* model is a model for this dataset of the form

$$Y_i \sim \text{Poisson}\left(\sum_{k=1}^m \beta_k e_{k,i}\right)$$

where $\beta_1, \dots, \beta_m$ are unknown parameters, assumed to be strictly positive.

- (i) Give pseudocode for fitting a Poisson-linear model. (ii) Express the model from part (a) as a Poisson-linear model. [4 marks]
- (c) In a factory, the workers suspect that one particular manager is incompetent, since there tend to be more safety incidents on days when that manager is present. They have assembled a dataset spanning several months, giving for each day the number of safety incidents and the names of the managers present. They believe that each manager has a competence level, and the number of incidents on a given day is related to the mean competence level among the managers present.
- (i) Suggest a probability model for this data, in the form of a Poisson-linear model. (ii) Explain how to test whether the suspect manager is indeed worse than the others. [11 marks]